034IN - FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL ADDITIONAL PARTIAL WRITTEN EXAMINATION A.Y. 2023/2024

September 2, 2024

First and Last Name:	
Group:	Group A
Exercise:	Exercise 1 - 2 Questions
Available Time Frame:	60 minutes

Important Notes:

- Write your answers on a single white sheet of paper using a black pen.
- Do not write in blue ink or pencil.
- Do not hand in additional sheets.
- The clarity and precision of your answers will be subject to evaluation.

I declare that the answers to this exercise are my own and only my own work and that I have not consulted with others.

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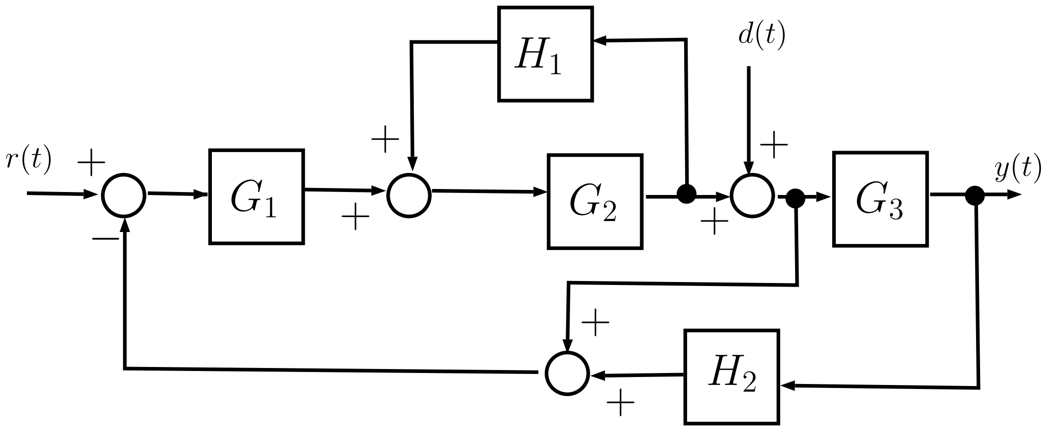
Exercise: Exercise 1

Available Time Frame: 60 minutes

Exercise 1

The closed-loop system shown in the block diagram has two input signals, namely $r(t)$ and $d(t)$, and one output signal, $y(t)$.



Question 1.1.

- Find the transfer functions $T_r^y(s)$, from the input $r(t)$ to the output $y(t)$, and T_d^y from the input $d(t)$ to the output $y(t)$.

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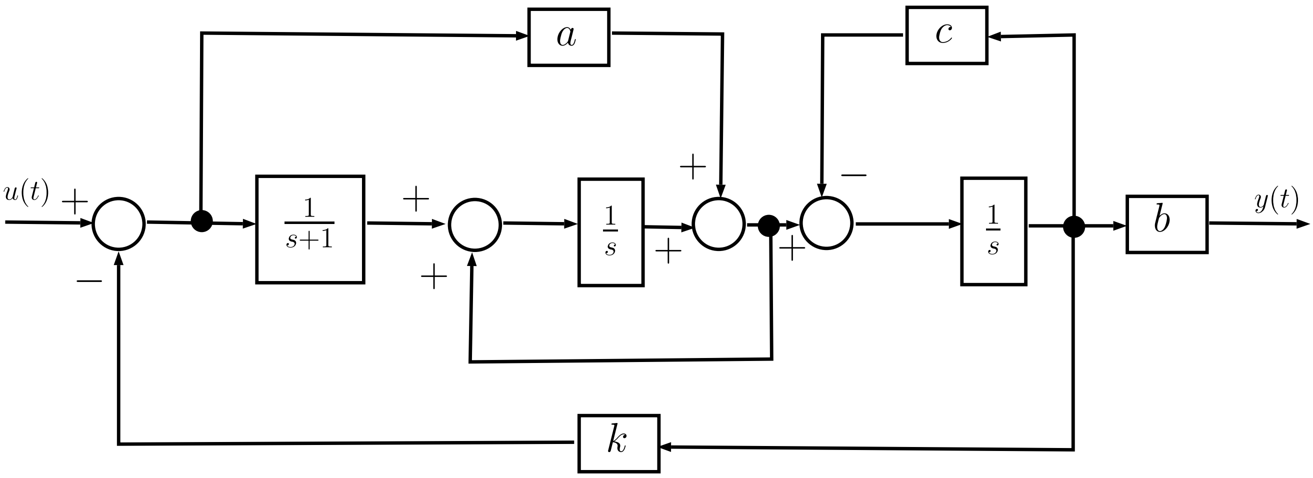
Exercise: Exercise 2

Available Time Frame: 60 minutes

Exercise 2

Consider the closed-loop system shown in the block diagram, with one input signal, $u(t)$, and one output signal, $y(t)$.



Assume the values

$$a = +3\,, \qquad b = +2\,, \qquad c = -3\,, \qquad k \in \mathbb{R}$$

for the proportional gain transfer functions in the diagram.

Question 2.1.

- Determine a state equations description for the dynamical system.
- Exploiting the state equations, analyse the stability of the system as a function of the parameter k.